Regression Analysis: Group Communication Apprehension and Public Transportation Usage

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Research Question

Does group communication apprehension correlate with public transportation usage?

Answer, Response, & Summary of Results

Yes, there is a correlational relationship between group communication apprehension and public transportation usage based on the merged survey data from Prolific demographic files and Qualtrics survey responses.

Merging Datasets

```
# Combine Prolific datasets
prolific_combined = pd.concat([prolific_a, prolific_b], ignore_index=True)

# Rename participant ID column for merging
prolific_combined = prolific_combined.rename(columns={'Participant id': 'participant_id'})
qualtrics = qualtrics.rename(columns={'Q0': 'participant_id'})

# Merge datasets on participant ID
merged_data = pd.merge(qualtrics, prolific_combined, on='participant_id', how='inner')

print(f"Merged dataset shape: {merged_data.shape}")
print(f"Number of participants: {merged_data['participant_id'].nunique()}")
```

```
Merged dataset shape: (252, 37)
Number of participants: 252
```

Variables

```
# Likert scale mapping (for communication apprehension)
likert_mapping = {
    'Strongly disagree': 1,
    'Somewhat disagree': 2,
    'Neither agree nor disagree': 3,
    'Somewhat agree': 4,
    'Strongly agree': 5
}

# transportation usage
usage_mapping = {
    'Never': 0,
    '0-1 days a month': 1,
```

```
    '2-4 days a month': 2,
    '4-8 days a month': 3,
    '8 or more days a month': 4
}

# Convert communication apprehension items (Q1-Q6, Q13-Q18) to numeric
ca_items = ['Q1', 'Q2', 'Q3', 'Q4', 'Q5', 'Q6', 'Q13', 'Q14', 'Q15', 'Q16', 'Q17', 'Q18']
reverse_coded = ['Q2', 'Q4', 'Q6', 'Q14', 'Q16', 'Q17']

for item in ca_items:
    if item in merged_data.columns:
        merged_data[item + '_num'] = merged_data[item].map(likert_mapping)
        # Reverse code items (1->5, 2->4, 3->3, 4->2, 5->1)
        if item in reverse_coded:
            merged_data[item + '_num'] = 6 - merged_data[item + '_num']

# Calculate communication apprehension mean score
ca_numeric_cols = [item + '_num' for item in ca_items if item + '_num' in merged_data.columns]
merged_data['ca_mean'] = merged_data[ca_numeric_cols].mean(axis=1, skipna=True)

# Convert transportation usage items (Q26-Q29) to numeric
pt_items = ['Q26', 'Q27', 'Q28', 'Q29']
for item in pt_items:
    if item in merged_data.columns:
        merged_data[item + '_num'] = merged_data[item].map(usage_mapping)

# Calculate public transportation usage mean score
pt_numeric_cols = [item + '_num' for item in pt_items if item + '_num' in merged_data.columns]
merged_data['pt_usage_mean'] = merged_data[pt_numeric_cols].mean(axis=1, skipna=True)

print(f"Communication apprehension mean: {merged_data['ca_mean'].mean():.2f}")
print(f"Public transportation usage mean: {merged_data['pt_usage_mean'].mean():.2f}")
```

```
Communication apprehension mean: 2.41
Public transportation usage mean: 1.82
```

Regression Analysis

Primary Question: Communication Apprehension predicting Public Transportation Usage

```
regression_data = merged_data[['ca_mean', 'pt_usage_mean']].dropna()

# regression model
if len(regression_data) > 0 and 'ca_mean' in regression_data.columns and 'pt_usage_mean' in regression_data
    X = regression_data['ca_mean'].values
    y = regression_data['pt_usage_mean'].values

    slope, intercept, r_value, p_value, std_err = stats.linregress(X, y)
    r_squared = r_value**2

    # results
    print("### Regression Results")
    print(f"R-squared: {r_squared:.3f}")
    print(f"Sample size: {len(y)}")
```

```python
    print(f"Slope: {slope:.4f}")
    print(f"Intercept: {intercept:.4f}")
    print(f"p-value: {p_value:.4f}")
    print(f"Standard error: {std_err:.4f}")
else:
    print("*Regression analysis requires actual data files to run properly.*")

table_data = {
    'R-squared': [f"{r_squared:.3f}"],
    'Sample size': [len(y)],
    'Slope': [f"{slope:.4f}"],
    'Intercept': [f"{intercept:.4f}"],
    'p-value': [f"{p_value:.4f}"],
    'Standard error': [f"{std_err:.4f}"]
}

pd.DataFrame(table_data)
```

```
### Regression Results
R-squared: 0.108
Sample size: 252
Slope: -0.4147
Intercept: 2.8160
p-value: 0.0000
Standard error: 0.0753
```

Table 1: Regression Results

	R-squared	Sample size	Slope	Intercept	p-value	Standard error
0	0.108	252	-0.4147	2.8160	0.0000	0.0753

Remaining Questions

- **Causality**: This analysis establishes correlation but cannot determine causality. Nature vs. Nurture: Does higher communication apprehension actually lead to reduced public transportation use, or do individuals with different transportation patterns develop different communication styles?
- **Measurement Validity**: The communication apprehension scale needs validation to ensure it reliably measures interest or comfortability in group settings.
- **Confounding Variables**: Additional control variables (urban/rural, income, accessibility needs, prevalence of transportation network) may influence individual's communication patterns and transportation choices.
- **Generalizability**: The sample may have selection bias from Prolific participants, which limits generalizability.